

Logical Model

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Relational Data Model

- Starting from the output of the Analysis of user requirements: **the Class Diagram**
- Generate the Logical Model by following a systematic process
 - organizing all data in flat **tables** of **rows** with a primary key
 - transforming associations as **foreign keys**

The diagram illustrates the transformation of class associations into foreign keys. It shows two tables: *Employees* and *Departments*. The *Employees* table has columns *Code*, *Name*, *Salary*, and *Dept*. The *Departments* table has columns *Code* and *Budget*. Arrows indicate the mapping of the *Dept* column in the *Employees* table to the *Code* column in the *Departments* table. Specifically, the values Y1, X2, and Y1 in the *Dept* column of the *Employees* table correspond to the values Y1, X2, and Y1 in the *Code* column of the *Departments* table.

<i>Employees</i>			
<i>Code</i>	<i>Name</i>	<i>Salary</i>	<i>Dept</i>
232	John	1000	Y1
143	Mary	1200	X2
254	Joan	900	Y1

<i>Departments</i>	
<i>Code</i>	<i>Budget</i>
Y1	100000
X2	750000

Algorithm for the Logical Design

- **Step I:** Translate all classes which are not involved in hierarchy
- **Step II:** Translate all hierarchies
- **Step III:** Translate multivalued attributes in to tables
- **Step IV :** Translate N-N relationships
- **Step V :** Translate 1-N relationships
- **Step VI :** Translate 1-1 relationships
- **Step VII :** Add other possible constraints

Step I: Translate all classes

- Classes become **tables** containing
 - Attributes
 - primary key
 - foreign keys
- **Primary key**
 - Simple to be used and compact
 - We can use set of attributes if compact
 - We can use a synthetic identifier

Students	T
studentID INTEGER	PK
Surname CHAR(20)	
Name CHAR(20)	
Year INTEGER	
Program CHAR(20)	
Supervisor CHAR(4)	FK

Teacher	T
code CHAR(4)	PK
...	



Step I: Translate all classes

Courses
code
title
program

Exams
mark
laud
date

Stages
Location
Date-start
Duration

Courses	T
Code CHAR(3)	PK
Title CHAR(20)	
Program CHAR(20)	

Identifier

Exams	T
Code CHAR(5)	PK
Mark INTEGER	
Laud BOOL	
Date DATE	

Synthetic
identifier

Stages	T
studentID INTEGER	PK, FK
Location CHAR(20)	
Date-start DATE	
Duration INTEGER	

External
Identifier

Step I: Translate all classes

Courses
code
title
program

Courses	T
Code CHAR(3)	PK
Title CHAR(20)	
Program CHAR(20)	

<u>Code</u>	Title	Program
PR1	Programming	Bachelor
ASD	Algorithms	Bachelor
DB1	Databases	Master

Step I: Translate all classes

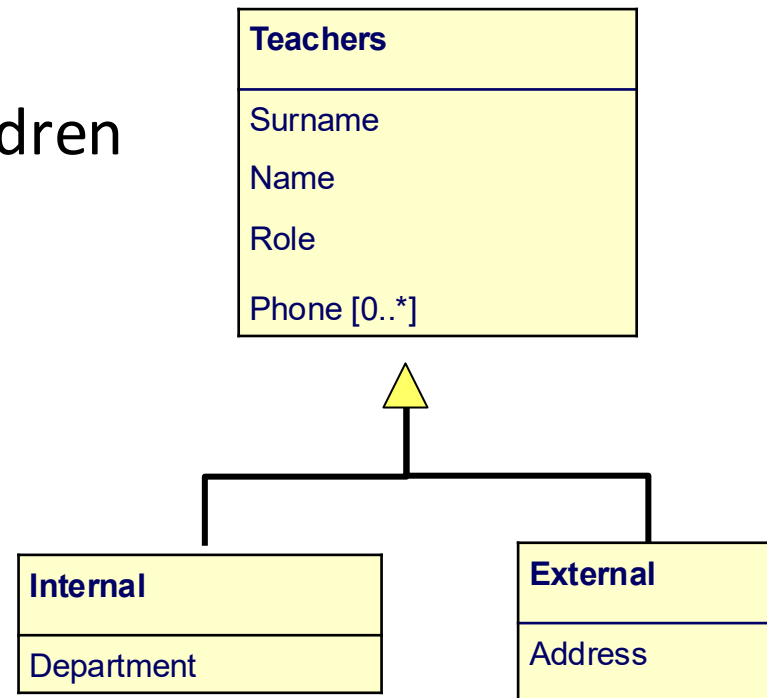
Stages
location
Date-start
duration

Stages	T
studentID INTEGER	PK, FK
Location CHAR(20)	
Date-start DATE	
Duration INTEGER	

<u>StudentID</u>	Location	Date-start	Duration
444	Microsoft	2002-05-15	3
77777	Microsoft	2002-05-15	3
88888	Basica	2002-09-01	3

Step II: Translate all hierarchies

- Relational Data Model does not have Hierarchies
- Three possible cases
 - Translate only the parent class
 - Translate only children
 - Translate both parent and children



Step II: Translate all hierarchies

- **Translate only the parent class**
 - One table with the name of the parent class
 - The table contains the attributes of both parent and children
 - Add an attribute **TYPE** to distinguish the instance
 - Generation of **NULL values**
 - It is convenient if the children **are not used so much** in the application

Teachers	T
Code CHAR(4)	PK
Surname CHAR(20)	NN
Name CHAR(20)	NN
Department CHAR(10)	
Role CHAR(15)	NN
Address CHAR(20)	
Type CHAR(10)	NN

Step II: Translate all hierarchies

- **Translate only children classes**
 - One table for each child
 - Each child table inherits both attributes and relationships of the parent
 - Possible only when the the **hierarchy is complete**
 - It is convenient if the application **uses often the children**

InternalTeachers	T
Code CHAR(4)	PK
Surname CHAR(20)	
Name CHAR(20)	
Department CHAR(10)	
Role CHAR(15)	

ExternalTeachers	T
Code CHAR(4)	PK
Surname CHAR(20)	
Name CHAR(20)	
Address CHAR(10)	
Role CHAR(15)	

Step II: Translate all hierarchies

- Translate both parent and children classes
 - One table for each child and for the parent
 - Each class has its attributes
 - It is necessary if the application **uses often both children and parent**

Teachers	T
Code CHAR(4)	PK
Surname CHAR(20)	
Name CHAR(20)	
Role CHAR(15)	

ExternalTeachers	T
Code CHAR(4)	PK,FK
Address CHAR(10)	

InternalTeachers	T
Code CHAR(4)	PK,FK
Department CHAR(10)	



Step III: Translate multivalued attributes

- Each multivalued attribute generates a new table
- The new table containing a foreign key vs the original class

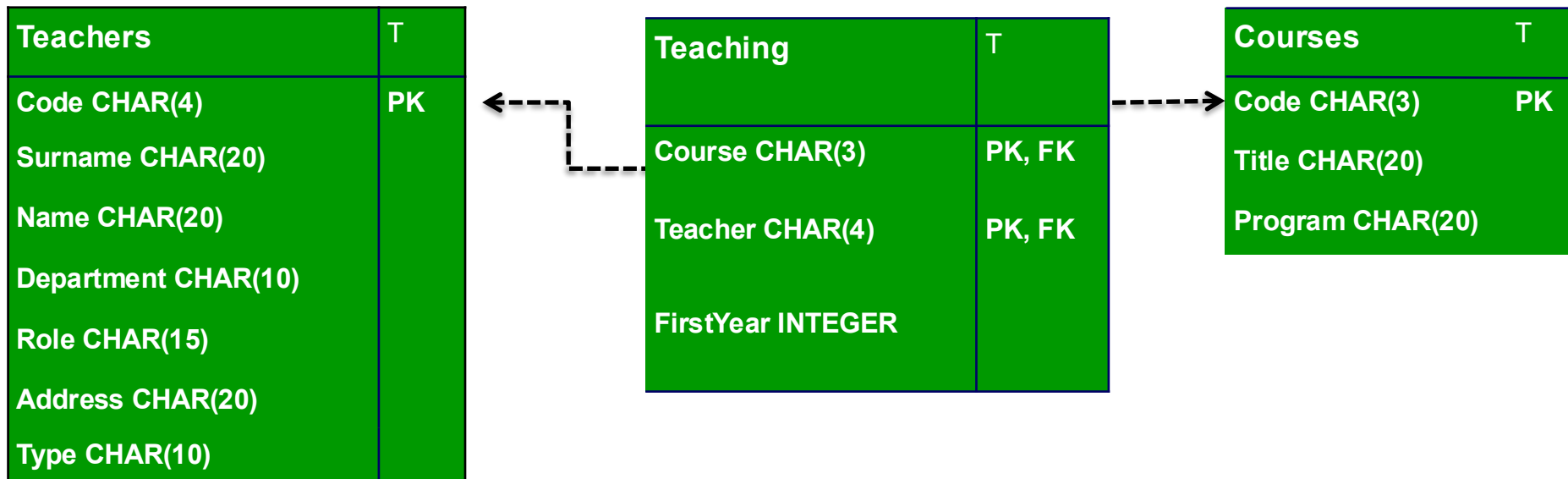
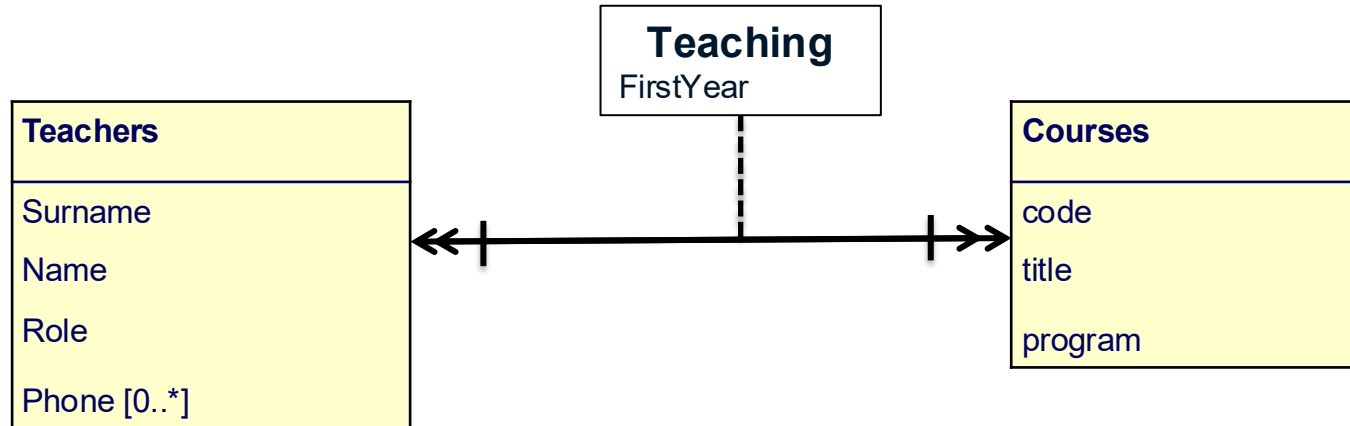
Teachers
Surname
Name
Role
Phone [0..*]

Teachers	T
Code CHAR(4)	PK
Surname CHAR(20)	
Name CHAR(20)	
Department CHAR(10)	
Role CHAR(15)	
Address CHAR(20)	
Type CHAR(10)	

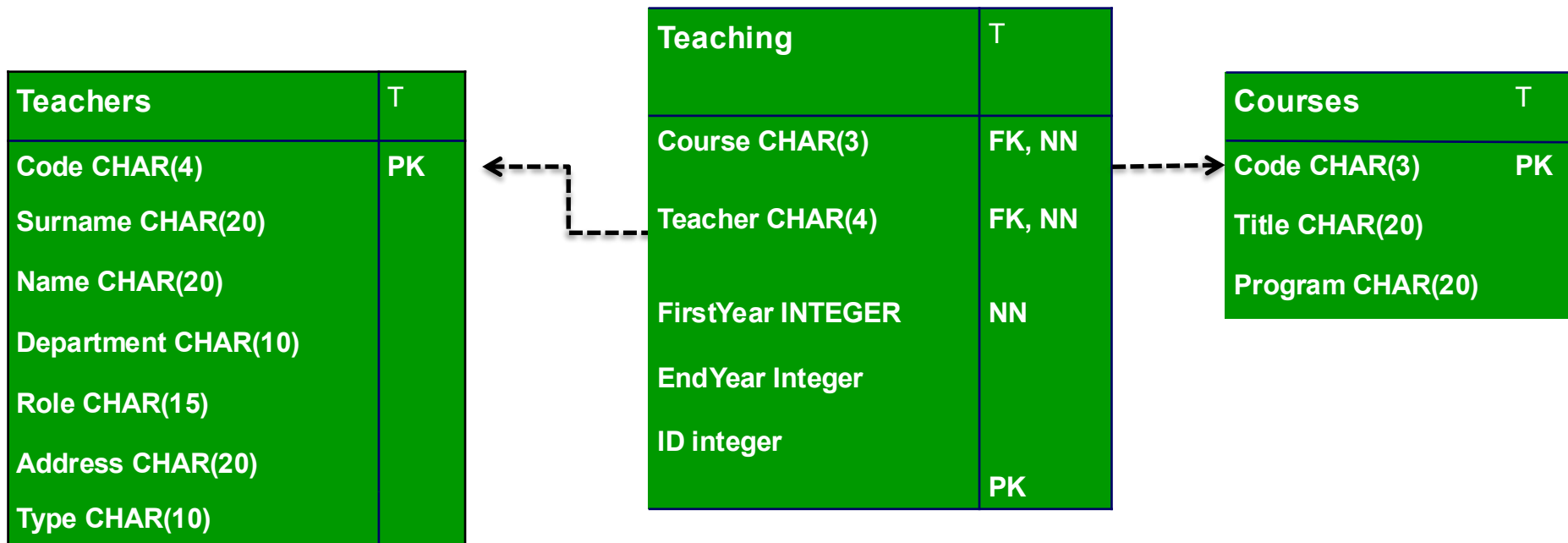
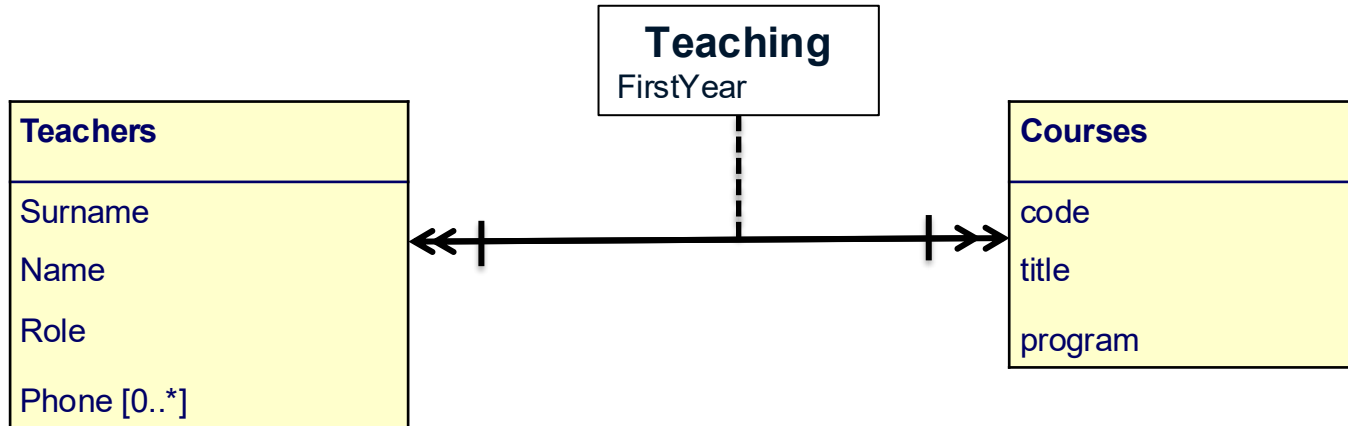
Phones	T
Number CHAR(15)	PK
Teacher CHAR(4)	FK



Step IV : Translate N-N relationships



Step IV : Translate N-N relationships



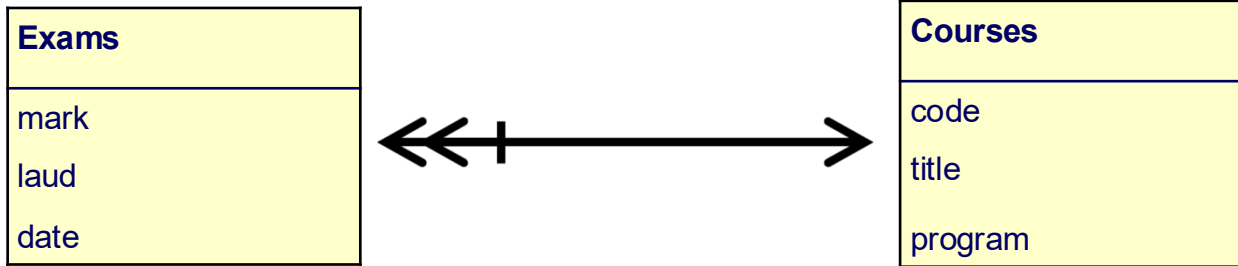
Step IV : Translate N-N relationships

<u>codice</u>	cognome	nome	...
FT	Pedreschi	Dino	...
CV	Monreale	Anna	...
ADP	Giannotti	Fosca	...

<u>docente</u>	<u>corso</u>	primoAnnoTit
FT	PR1	2001
CV	ASD	2002
FT	ASD	1999
...	...	

<u>Code</u>	Title	Program
PR1	Programming	Bachelor
ASD	Algorithms	Bachelor
DB1	Databases	Master

Step V : Translate 1-N relationships



Exams	T
Code CHAR(5)	PK
Course CHAR(3)	FK
Mark INTEGER	
Laud BOOL	
Date DATE	

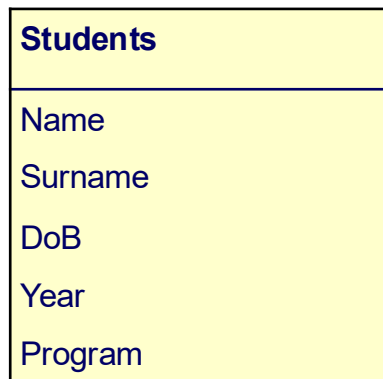
Courses	T
Code CHAR(3)	PK
Title CHAR(20)	
Program CHAR(20)	

<u>Code</u>	Title	Program
PR1	Programming	Bachelor
ASD	Algorithms	Bachelor
DB1	Databases	Master

<u>Code</u>	Course	Mark	Laud	Date
pr101	PR1	27	false	2002-06-12
asd01	ASD	30	true	2001-12-03
BD101	INFT	24	false	2001-09-30
pr102	PR1	21	false	2002-06-12
asd02	ASD	20	false	2001-12-03
asd03	ASD	28	false	2002-06-13

Step VI : Translate 1-1 relationships

- Similar to 1-N relationship but we can choose where putting the foreign key
- It is better in the class where we have total relationship



Students	T
studentID INTEGER	PK
Surname CHAR(20)	
Name CHAR(20)	
Year INTEGER	
Program CHAR(20)	
Supervisor CHAR(4)	FK

Stages	T
studentID INTEGER	PK, FK
Location CHAR(20)	
Date-start DATE	
Duration INTEGER	



Step VII: Add other possible constraints

- Now we have:
 - Tables
 - Attributes
 - Primary keys
 - Foreign keys
- We need to add other constraints
 - NOT NULL
 - DEFAULT
 - CASCADE
 - CHECK
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